

Alirocumab in Pediatric Patients With Heterozygous Familial Hypercholesterolemia

A Randomized Clinical Trial

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 Supplemental content

IMPORTANCE Many pediatric patients with heterozygous familial hypercholesterolemia (HeFH) cannot reach recommended low-density lipoprotein cholesterol (LDL-C) concentrations on statins alone and require adjunct lipid-lowering therapy (LLT); the use of alirocumab in pediatric patients requires evaluation.

OBJECTIVE To assess the efficacy of alirocumab in pediatric patients with inadequately controlled HeFH.

DESIGN, SETTING, AND PARTICIPANTS This was a phase 3, randomized clinical trial conducted between May 2018 and August 2022 at 43 centers in 24 countries. Pediatric patients aged 8 to 17 years with HeFH, LDL-C 130 mg/dL or greater, and receiving statins or other LLTs were included. Following consecutive enrollment into dosing cohorts, 25 of 99 patients screened for dosing every 2 weeks (Q2W) failed screening; 25 of 104 patients screened for dosing every 4 weeks (Q4W) failed screening. A total of 70 of 74 Q2W patients (95%) and 75 of 79 Q4W patients (95%) completed the double-blind period.

INTERVENTIONS Patients were randomized 2:1 to subcutaneous alirocumab or placebo and Q2W or Q4W. Dosage was based on weight (40 mg for Q2W or 150 mg for Q4W if <50 kg; 75 mg for Q2W or 300 mg for Q4W if ≥50 kg) and adjusted at week 12 if LDL-C was 110 mg/dL or greater at week 8. After the 24-week double-blind period, patients could receive alirocumab in an 80-week open-label period.

MAIN OUTCOMES AND MEASURES The primary end point was percent change in LDL-C from baseline to week 24 in each cohort.

RESULTS Among 153 patients randomized to receive alirocumab or placebo (mean [range] age, 12.9 [8-17] years; 87 [56.9%] female), alirocumab showed statistically significant reductions in LDL-C vs placebo in both cohorts at week 24. Least squares mean difference in percentage change from baseline was -43.3% (97.5% CI, -56.0 to -30.7; $P < .001$) Q2W and -33.8% (97.5% CI, -46.4 to -21.2; $P < .001$) Q4W. Hierarchical analysis of secondary efficacy end points demonstrated significant improvements in other lipid parameters at weeks 12 and 24 with alirocumab. Two patients receiving alirocumab Q4W experienced adverse events leading to discontinuation. No significant difference in adverse event incidence was observed between treatment groups. Open-label period findings were consistent with the double-blind period.

CONCLUSIONS AND RELEVANCE The findings in this study indicate that alirocumab Q2W or Q4W significantly may be useful for reducing LDL-C and other lipid parameters and be well tolerated in pediatric patients with HeFH inadequately controlled with statins.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: [NCT03510884](https://clinicaltrials.gov/ct2/show/study/NCT03510884)

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Heterozygous familial hypercholesterolemia (HeFH) is an autosomal dominant disorder affecting approximately 1 in 313 individuals characterized by severely elevated low-density lipoprotein cholesterol (LDL-C) concentrations.^{1,2} If left untreated, patients may develop early-onset atherosclerotic cardiovascular disease. Detection and intervention to control LDL-C concentrations in pediatric patients can prevent atherosclerosis progression and premature cardiovascular events in adulthood.^{3,4} American College of Cardiology/American Heart Association guidelines define acceptable LDL-C concentrations in pediatric and adolescent patients as less than 110 mg/dL (to convert to mmol/L, multiply by 0.0259) and recommend initiation of statins (or nonstatins if statin intolerant) from as young as 8 years.^{5,6} Current European Atherosclerosis Society pediatric guidelines recommend LDL-C reduction to less than 130 mg/dL for patients over 10 years, or by at least 50% for patients from 8 years.⁷ Other lipid-lowering therapies (LLTs), including ezetimibe and bile-acid sequestrants, are available as adjuncts to statins or as alternative therapy for those who are statin intolerant.⁵

However, some pediatric patients cannot achieve LDL-C targets despite high-dose statins or multiagent LLTs. Additional LLTs are needed for these patients to help control LDL-C concentrations and reduce risk of atherosclerotic disease in adulthood. Recent therapeutic developments include inhibitors of proprotein convertase subtilisin kexin type 9 (PCSK9), a protease that promotes degradation of the low-density lipoprotein receptor responsible for clearing excess LDL-C from the blood.^{8,9}

Alirocumab is a fully human monoclonal antibody that binds and inhibits circulating PCSK9. Alirocumab has been shown to reduce LDL-C and other atherogenic lipoproteins in adults with hyperlipidemia—including patients with HeFH—either alone or in combination with other LLTs. Alirocumab has shown to reduce cardiovascular event risk in adult patients with established cardiovascular disease.^{10,11} In adults in the US, the starting dose for alirocumab is 75 mg once every 2 weeks (Q2W) or 300 mg once every 4 weeks (Q4W); in the European Union, the starting dose is 75 mg Q2W, and for patients requiring LDL-C reduction of greater than 60% the starting dose is 150 mg Q2W or 300 mg Q4W.¹²⁻¹⁴

A phase 2 dose-finding study, the 8-Week Open-Label, Sequential, Repeated Dose-Finding Study to Evaluate the Efficacy and Safety of Alirocumab in Children and Adolescents With Heterozygous Familial Hypercholesterolemia Followed by an Extension Phase (ODYSSEY KIDS),⁸ evaluated alirocumab in pediatric patients with HeFH inadequately controlled with statins. Alirocumab was well tolerated and reduced LDL-C with results at higher doses (−46.1% at week 8 for 40 mg/75 mg Q2W if <50 kg/≥50 kg), consistent with reductions reported in adults, supporting further investigation in pediatric HeFH. This phase 3 trial assessed the efficacy of alirocumab in pediatric patients with HeFH inadequately controlled with statins.

Methods

Study Design

This was a phase 3, randomized, double-blind, placebo-controlled, multicenter clinical trial. Patients were screened

Key Points

Question Is alirocumab an efficacious lipid-lowering therapy (LLT) for pediatric patients (8-17 years) with heterozygous familial hypercholesterolemia (HeFH) inadequately controlled with statins?

Findings In this randomized clinical trial of 153 patients, alirocumab treatment showed statistically significant reductions in low-density lipoprotein cholesterol (LDL-C) vs placebo in 2 dosing cohorts.

Meaning The findings indicate that alirocumab may be an effective adjunct LLT for pediatric patients with HeFH inadequately controlled with statins.

at 43 centers in 24 countries (Argentina, Austria, Brazil, Bulgaria, Canada, Czech Republic, Denmark, Finland, France, Hungary, Italy, Lebanon, Mexico, the Netherlands, Norway, Poland, Russia, Slovenia, South Africa, Spain, Sweden, Taiwan, Turkey, and the US), with 40 centers randomizing patients. The study was conducted in accordance with the protocol (Supplement 1) and international guidelines, including the Declaration of Helsinki and the International Conference on Harmonization guidelines for Good Clinical Practice, and all applicable local laws and regulations. The study protocol was approved by the independent review boards or ethics committees of participating centers. Signed parental consent, with or without patient consent, was obtained for all patients. The study is registered at ClinicalTrials.gov (NCT03510884). The study protocol and the statistical analysis plan are available in Supplement 1.

Patient Eligibility Criteria

Patients aged 8 to 17 years were eligible if they had inadequately controlled HeFH (LDL-C ≥130 mg/dL) despite treatment with optimal statin dose (or nonstatin LLT, if statin intolerant), with or without other LLT, for at least 4 weeks. HeFH diagnosis was confirmed by genotyping or Simon Broome criteria.¹⁵ Optimal statin dose was defined as a stable daily dose prescribed based on regional or local guidelines or the maximally tolerated dose. Statin intolerance was defined as the inability to tolerate at least 2 statins (1 at the lowest daily dose, the others at any dose) or patients not receiving a daily statin. Patients who had participated in ODYSSEY KIDS⁸ were eligible and not excluded based on LDL-C less than 130 mg/dL, as LDL-C 130 mg/dL or greater was required at screening for ODYSSEY KIDS.

Patients were excluded if they were less than 25 kg in body weight, aged 8 to 9 years and not at Tanner Stage 1, or aged 10 to 17 years and not at Tanner Stage 2 or higher. Other exclusion criteria included secondary hyperlipidemia; homozygous FH; prior lipid apheresis within 2 months of screening or planned apheresis during the study; uncontrolled type 1 or 2 diabetes (per local guidelines), thyroid disease or hypertension; severe kidney impairment; abnormal liver function; or creatine phosphokinase greater than 3 times the upper limit of normal.

Study Interventions

The study included a 4-week run-in period and a 2-week screening period; ODYSSEY KIDS⁸ participants were required to have

a washout of at least 10 weeks between their last alirocumab dose and screening lipid assessment. Patients were randomized 2:1 in 2 consecutive dosing cohorts (Q2W or Q4W) to receive alirocumab or matching placebo for 24 weeks in addition to their stable background LLT. Randomization was stratified by previous participation in ODYSSEY KIDS⁸ and baseline weight. In the Q2W cohort, alirocumab dose was 40 mg for patients weighing less than 50 kg and 75 mg for those weighing 50 kg or more. In the Q4W cohort, the alirocumab dose was 150 mg for patients weighing less than 50 kg and 300 mg for those weighing 50 kg or more. If LDL-C at week 8 was 110 mg/dL or higher, the alirocumab dose was adjusted at week 12 to 75 mg Q2W for patients weighing less than 50 kg and 150 mg Q2W for those weighing 50 kg or more, regardless of cohort. Lipid values during the 24-week double-blind period were not communicated to the investigator; continuation of dose or modification at week 12 was a blinded and automated process. Dosing by weight was matched in the placebo arms (Supplement 1).

After completing 24 weeks of double-blind treatment, all patients could enter an open-label period to receive alirocumab in addition to their LLT for up to 80 weeks. The dose was adjusted based on investigator judgment (taking LDL-C and weight into consideration) (Supplement 1).

Assessments

Blood samples monitoring LDL-C concentrations, other lipids, and clinical laboratory parameters were collected at baseline and weeks 8, 12, and 24 of the double-blind period, and at weeks 32, 44, 68, and 104 (study completion) of the open-label period. Assessment of efficacy end points were handled by a central laboratory. Safety was assessed throughout the study and included adverse event (AE) reports, vital signs, weight, and height measurements, Cogstate battery test score, Tanner stage, laboratory data, and antialirocumab antibodies.

Study End Points

The primary end point was the percent change in LDL-C from baseline to week 24 in each cohort. Key secondary efficacy end points were percent change in LDL-C at week 12 and other lipid parameters at weeks 12 and 24, including: apolipoprotein B (apo B), non-high-density lipoprotein cholesterol (non-HDL-C), total cholesterol, high-density lipoprotein cholesterol (HDL-C), lipoprotein (a) (Lp[a]), triglycerides, and apolipoprotein A-I. Safety, tolerability, and development of antialirocumab antibodies after 24 weeks were also secondary end points.

Statistical Analysis

The trial planned for a sample size of 90 patients to provide 92% power in identifying a difference in mean percent change in LDL-C of 30% between alirocumab and placebo. The size was increased to 150 patients to properly assess safety and tolerability.

The primary efficacy end point was analyzed in the intention-to-treat population using a mixed-effects model with repeated-measures approach; the model included treatment group, randomization strata, time points, treatment-by-time point interaction, strata-by-time point interaction, continuous fixed covariates of baseline value, and baseline value-by-

time point interaction. Parameters were estimated using restricted maximum likelihood method with the Newton-Raphson algorithm. To address multiple testing, a Bonferroni adjustment was applied, using a 2-sided α level of .025 within each cohort. Due to the sequential enrollment of participants in 2 different cohorts, comparisons were made between treatment groups within each dosing cohort separately.

Continuous secondary efficacy end points anticipated to have a normal distribution were analyzed using the same approach as the primary end point; those anticipated to have a nonnormal distribution were analyzed using a multiple imputation approach followed by a robust regression model using M-estimation with treatment group and randomization strata as main effects and corresponding baseline value as covariate. Categorical secondary efficacy end points were analyzed using a multiple imputation approach followed by logistic regression model adjusting for treatment group as main effect and corresponding baseline value as covariate, stratified by randomization factors. To address multiple testing, key secondary efficacy end points were analyzed in each cohort if the primary end point was met and using a hierarchical testing procedure with a 2-sided α of .05.

Efficacy analyses were performed on the intention-to-treat population, defined as all randomized patients. For analyses of the Q4W cohort, participation in ODYSSEY KIDS was not included in randomization strata because there were too few patients. Efficacy data for the open-label period were summarized descriptively.

Safety data were summarized with descriptive statistics for the safety population of the double-blind period and the open-label population. The safety population consisted of all randomized patients who received at least 1 dose or partial dose of study treatment, analyzed for the actual treatment received.

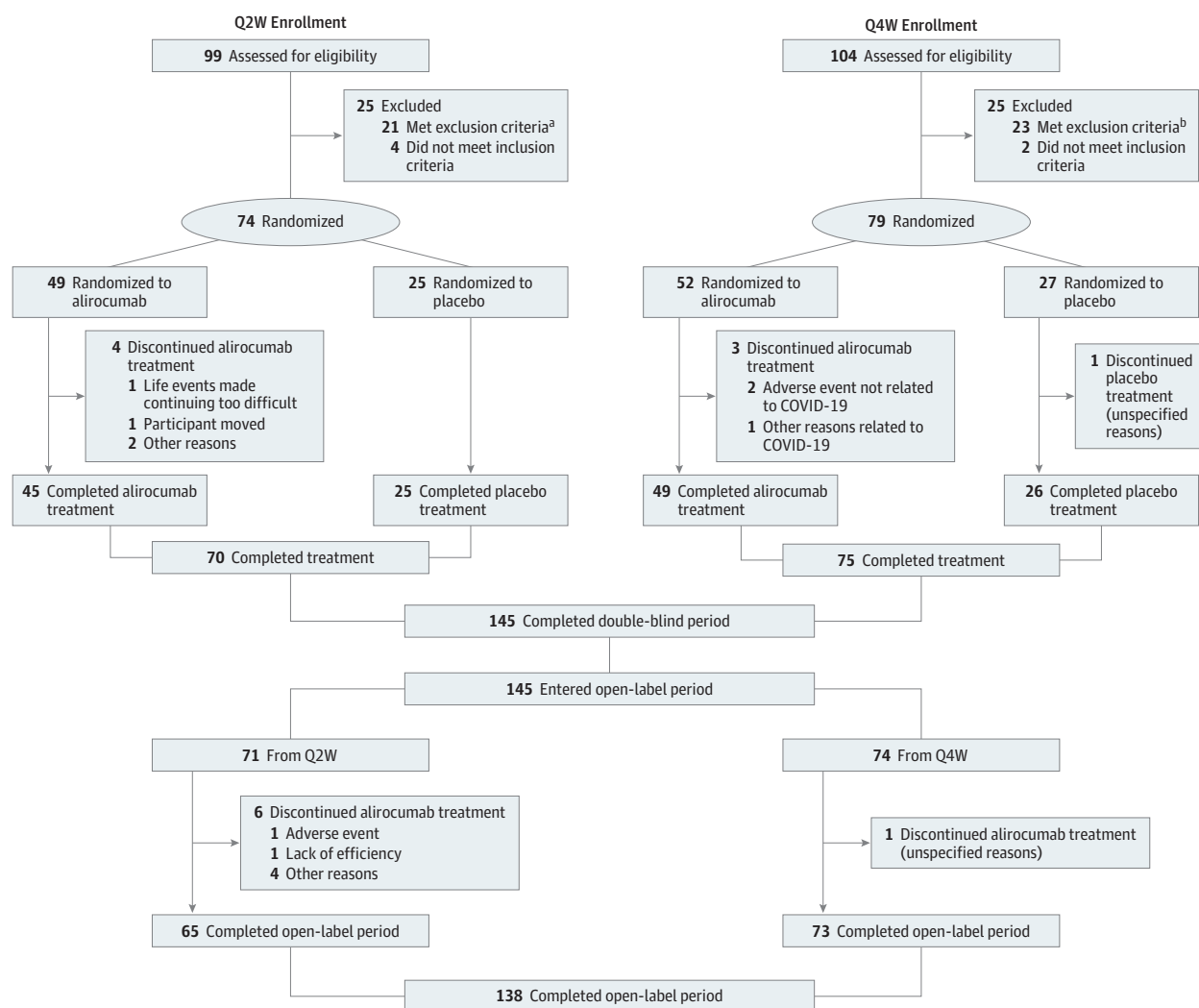
Results

Patients

A total of 153 patients (mean [range] age, 12.9 [8-17] years; 87 [56.9%] female) were randomized 2:1 to alirocumab or placebo, following Q2W ($n = 74$; 49 alirocumab and 25 placebo) or Q4W ($n = 79$; 52 alirocumab and 27 placebo) dosing. At week 12, alirocumab doses were up-titrated in 22 patients (44.9%) in Q2W and 15 (28.8%) in Q4W. The first patient visit was on May 31, 2018, and the last patient visit on August 5, 2022. Seventy patients (95%) in the Q2W cohort and 75 (95%) in Q4W completed the double-blind treatment period. Subsequently, 145 patients entered and 138 (95%) completed the open-label period (Figure 1).

Baseline patient demographic characteristics, disease characteristics, and lipid parameters were broadly similar between treatment groups and dosing cohorts, with some exceptions (Table 1). In the Q2W cohort, there were more patients weighing less than 50 kg, more ODYSSEY KIDS participants, and fewer female participants than in the Q4W cohort. At baseline, LDL-C concentrations were similar between treatment groups in the Q2W and Q4W cohorts.

Figure 1. Participant Flow Diagram



Q2W indicates dosing once every 2 weeks; Q4W, dosing once every 4 weeks.

^a Includes 12 patients (12.1%) with low-density lipoprotein cholesterol less than 130 mg/dL (to convert to mmol/L, multiply by 0.0259) after the patient had been receiving stable lipid-modifying treatment for at least 4 weeks; 5 patients (5.1%) whose parents withdrew consent during the screening period; 3 patients (3.0%) aged 8 to 9 years not at Tanner stage 1 or aged 10 to 17 years not at Tanner stage 2 or higher; 1 patient (1.0%) with uncontrolled type 1 or 2 diabetes (ie, hemoglobin A_{1c} levels above local guidelines or equivalent); and 1 patient (1.0%) with creatine phosphokinase 3 times the upper limit of

normal or higher.

^b Includes 14 patients (13.5%) with low-density lipoprotein cholesterol less than 130 mg/dL after the patient had been receiving stable lipid-modifying treatment for at least 4 weeks; 4 patients (3.8%) with any abnormal clinical condition or disease during screening that would hinder study completion or any other condition in the investigator's opinion as making the patient inappropriate for the study; 3 patients (2.9%) whose parents withdrew consent during the screening period; and 2 patients (1.9%) aged 8 to 9 years not at Tanner stage 1 or aged 10 to 17 years not at Tanner stage 2 or higher.

Efficacy

Alirocumab was associated with LDL-C reductions from baseline at first assessment (week 8) of the double-blind period in both cohorts, with reductions maintained through week 24 (Figure 2). For the primary end point, alicumab significantly reduced LDL-C compared with placebo in both dosing cohorts at week 24, with a least squares (LS) mean difference of −43.3% (97.5% CI, −56.0 to −30.7; $P < .001$) in Q2W and −33.8% (97.5% CI, −46.4 to −21.2; $P < .001$) in Q4W. The LDL-C reduction associated with alicumab at week 24 was maintained across all subgroups of interest, including those defined

by baseline weight, sex, age, and LDL-C concentration (eFigure 1 in Supplement 2).

At baseline, free PCSK9 concentrations were similar across treatment groups (Table 1). At week 24, mean (SD; IQR) free PCSK9 was 17.4 (54.0; 0.0–0.0) ng/mL (−86.0% from baseline) for alicumab Q2W and 19.0 (42.4; 0.0–0.0) ng/mL (−88.8%) for alicumab Q4W, compared with 180.4 (66.5; 121.0–240.0) ng/mL (−5.5%) for placebo Q2W and 104.8 (52.4; 57.0–136.0) ng/mL (−17.1%) for placebo Q4W.

Alirocumab was associated with significant reductions vs placebo through hierarchical testing of the first 12 key

Table 1. Baseline Characteristics of the Intention-to-Treat Population

Characteristic	No. (%)					
	Q2W cohort (n = 74)		Q4W cohort (n = 79)		Overall (N = 153)	
	Alirocumab (n = 49)	Placebo (n = 25)	Alirocumab (n = 52)	Placebo (n = 27)	Q2W (n = 74)	Q4W (n = 79)
Age, mean (SD), y	12.5 (2.7)	13.2 (2.4)	13.1 (3.0)	12.8 (3.0)	12.8 (2.6)	13.0 (3.0)
<12 y	19 (38.8)	6 (24.0)	20 (38.5)	10 (37.0)	25 (33.8)	30 (38.0)
≥12 y	30 (61.2)	19 (76.0)	32 (61.5)	17 (63.0)	49 (66.2)	49 (62.0)
Sex						
Female	30 (61.2)	8 (32.0)	34 (65.4)	15 (55.6)	38 (51.4)	49 (62.0)
Male	19 (38.8)	17 (68.0)	18 (34.6)	12 (44.4)	36 (48.6)	30 (38.0)
Race ^a						
White	42 (85.7)	23 (92.0)	38 (73.1)	22 (81.5)	65 (87.8)	60 (75.9)
Other ^b	7 (14.3)	2 (8.0)	14 (26.9)	5 (18.5)	9 (12.2)	19 (24.1)
Weight, mean (SD), kg	53.9 (22.2)	50.4 (15.1)	54.7 (20.2)	50.9 (14.0)	52.7 (20.1)	53.4 (18.3)
<50 kg	25 (51.0)	13 (52.0)	20 (38.5)	10 (37.0)	38 (51.4)	30 (38.0)
≥50 kg	24 (49.0)	12 (48.0)	32 (61.5)	17 (63.0)	36 (48.6)	49 (62.0)
Height, mean (SD), cm	156.2 (16.5)	156.6 (14.0)	155.6 (16.5)	154.2 (13.0)	156.3 (15.6)	155.1 (15.3)
ODYSSEY KIDS study participation	19 (38.8)	10 (40.0)	2 (3.8)	1 (3.7)	29 (39.2)	3 (3.8)
Time to diagnosis, mean (SD), y	4.21 (3.11)	4.35 (3.76)	2.45 (2.66)	3.71 (3.80)	4.26 (3.32)	2.88 (3.13)
Diagnosis of HeFH						
SB criteria	29 (59.2)	14 (56.0)	22 (42.3)	11 (40.7)	43 (100)	33 (100)
Genotyping prior to or during the study	43 (87.8)	19 (76.0)	50 (96.2)	26 (96.3)	62 (83.8)	76 (96.2)
Cardiovascular disease history in the family/cardiovascular risk factors	48 (98.0)	25 (100)	51 (98.1)	26 (96.3)	73 (98.6)	77 (97.5)
Never smoked	49 (100)	25 (100)	51 (98.1)	26 (96.3)	74 (100)	77 (97.5)
Baseline lipid parameters						
LDL-C, mean (SD), mg/dL	169.69 (46.74)	175.29 (50.23)	176.79 (53.93)	176.57 (49.01)	171.58 (47.68)	176.71 (51.99)
Apolipoprotein B, mean (SD), mg/dL	115.7 (24.5)	115.2 (25.8)	119.7 (29.3)	118.4 (31.2)	115.5 (24.8)	119.2 (29.8)
Non-HDL-C, mean (SD), mg/dL	186.75 (48.29)	191.61 (50.64)	197.16 (55.14)	195.37 (53.98)	188.39 (48.81)	196.55 (54.41)
Total cholesterol, mean (SD), mg/dL	234.69 (49.65)	242.88 (59.06)	246.98 (57.66)	249.74 (53.49)	237.46 (52.75)	247.92 (55.94)
HDL-C, mean (SD), mg/dL	48.24 (10.61)	51.16 (13.07)	49.62 (11.06)	54.74 (14.20)	49.23 (11.50)	51.37 (12.38)
HDL-C ≥40 mg/dL	38 (77.6)	22 (88.0)	42 (80.8)	23 (85.2)	60 (81.1)	65 (82.3)
Fasting triglyceride, median (IQR), mg/dL	65.00 (54.00-89.38)	68.14 (51.33-85.84)	91.00 (72.57-110.62)	82.00 (52.98-125.66)	66.57 (51.33-89.38)	87.17 (61.95-116.81)
Lipoprotein(a), median (IQR), mg/dL	13.2 (2.0-44.1)	22.0 (6.3-69.4)	12.7 (5.6-35.0)	17.8 (5.5-43.4)	16.2 (5.2; 54.7)	13.9 (5.5-40.9)
Free PCSK9, mean (SD), ng/mL	188.4 (89.2)	212.1 (79.9)	168.0 (87.2)	158.8 (76.5)	NA	NA
Baseline treatment						
Statin intolerant	2 (4.1)	1 (4.0)	5 (9.6)	3 (11.1)	3 (4.1)	8 (10.1)
Ezetimibe	2 (4.1)	3 (12.0)	12 (23.1)	4 (14.8)	5 (6.8)	16 (20.3)
Any statin	49 (100)	24 (96.0)	48 (92.3)	24 (88.9)	73 (98.6)	72 (91.1)
Atorvastatin, mg						
<10	2 (4.1)	2 (8.0)	3 (5.8)	3 (11.1)	4 (5.4)	6 (7.6)
10	22 (44.9)	6 (24.0)	6 (11.5)	5 (18.5)	28 (37.8)	11 (13.9)
20	6 (12.2)	8 (32.0)	8 (15.4)	5 (18.5)	14 (18.9)	13 (16.5)
40	0	0	13 (25.0)	4 (14.8)	0	17 (21.5)
80	0	0	0	0	0	0
Other dose	1 (2.0)	0	1 (1.9)	0	1 (1.4)	1 (1.3)
Rosuvastatin, mg						
<5	1 (2.0)	0	0	0	1 (1.4)	0
5	3 (6.1)	0	2 (3.8)	2 (7.4)	3 (4.1)	4 (5.1)
10	6 (12.2)	1 (4.0)	4 (7.7)	1 (3.7)	7 (9.5)	5 (6.3)
20	0	2 (8.0)	1 (1.9)	1 (3.7)	2 (2.7)	2 (2.5)
40	0	0	2 (3.8)	0	0	2 (2.5)
Other dose	0	1 (4.0)	2 (3.8)	1 (3.7)	1 (1.4)	4 (3.8)

(continued)

Table 1. Baseline Characteristics of the Intention-to-Treat Population (continued)

Characteristic	No. (%)					
	Q2W cohort (n = 74)		Q4W cohort (n = 79)		Overall (N = 153)	
	Alirocumab (n = 49)	Placebo (n = 25)	Alirocumab (n = 52)	Placebo (n = 27)	Q2W (n = 74)	Q4W (n = 79)
Pravastatin, mg						
<10	0	0	0	0	0	0
10	1 (2.0)	2 (8.0)	1 (1.9)	0	3 (4.1)	1 (1.3)
20	3 (6.1)	1 (4.0)	3 (5.8)	1 (3.7)	4 (5.4)	4 (5.1)
40	1 (2.0)	0	0	0	1 (1.4)	0
>40	0	0	0	0	0	0
Other dose	1 (2.0)	0	0	0	1 (1.4)	0

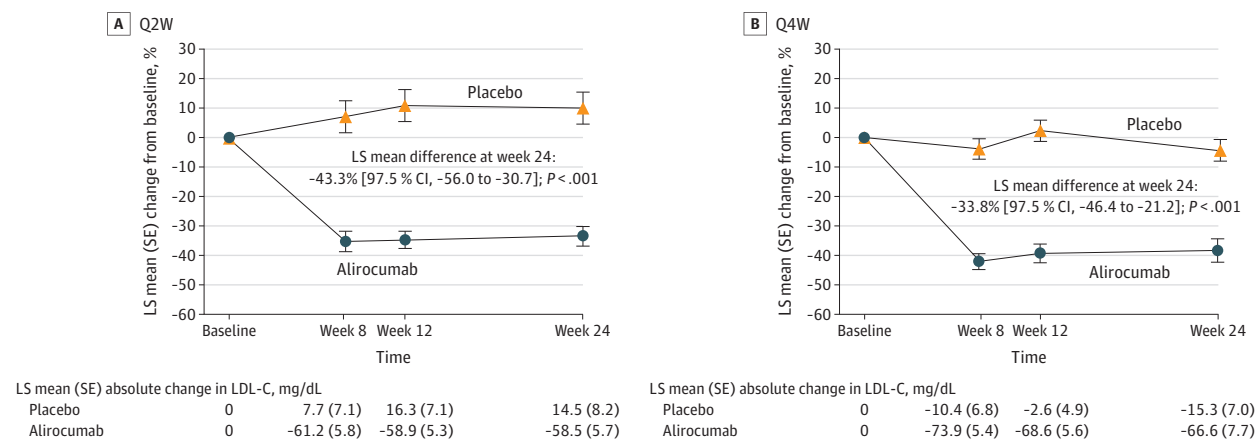
Abbreviations: HDL-C, high-density lipoprotein cholesterol; HeFH, familial heterozygous hypercholesterolemia; LDL-C, low-density lipoprotein cholesterol; Q2W, dosing every 2 weeks; Q4W, dosing every 4 weeks; SB, Simon Broome.

^a Race and ethnicity data were collected via questionnaire and reported because these data are required by the US Food and Drug Administration

and the European Medicines Agency.

^b To preserve anonymity, racial categories with small numbers of patients have been grouped together: American Indian or Alaska Native, Asian, Black, Black and multiple races, Native Hawaiian or Other Pacific Islander, Native Hawaiian or Other Pacific Islander and White, and not reported.

Figure 2. Change in Low-Density Lipoprotein Cholesterol (LDL-C) From Baseline in the Intention-to-Treat Population



LS means and SEs taken from mixed-effect model with repeated-measures analysis. The model includes the fixed categorical effects of treatment group, stratum previous participation (yes or no) in the ODYSSEY KIDS study (except for Q4W cohort), baseline body weight (<50 or ≥50 kg), time point (week 8,

week 12, and week 24), treatment-by-time point interaction, and strata-by-time point interaction, as well as the continuous fixed covariates of baseline LDL-C value and baseline LDL-C value-by-time point interaction. LS indicates least squares; Q2W, dosing once every 2 weeks; Q4W, dosing once every 4 weeks.

secondary efficacy end points, including apo B, non-HDL-C, and total cholesterol. At week 24, there was a -15.2% (97.5% CI, -30.3 to -0.1) reduction in the adjusted mean difference of Lp(a) in the Q2W cohort and a -24.9% (97.5% CI, -44.4 to -5.4) reduction in Q4W (Table 2). At week 24, 38 patients (77.3%) receiving alirocumab in the Q2W cohort and 39 (76.3%) in Q4W achieved LDL-C less than 130 mg/dL, and 10 (21.6%) in Q2W and 17 (32.4%) in Q4W achieved LDL-C reduction of 50% or greater.

In the open-label period, LDL-C reductions were well maintained in patients who received alirocumab during the double-blind period, while those who switched from placebo to alirocumab had reductions in LDL-C at week 32 that were maintained through week 104. In the Q2W cohort, the LS mean (SD) change in LDL-C from baseline to week 104 was -26.3% (28.0) and -23.9% (33.5) in the Q4W cohort. At week 104, 45 (63.7%) in the Q2W cohort and 44 (59.6%) in the Q4W cohort

achieved LDL-C less than 130 mg/dL, while 10 (13.8%) in Q2W and 15 (21.1%) in Q4W achieved ≥50% LDL-C reduction.

Safety

During the double-blind period, AEs occurred in 26 patients (53.1%) receiving alirocumab and 13 (52.0%) receiving placebo in the Q2W cohort and 26 (50.0%) and 16 (59.3%), respectively, in Q4W (Table 3). Most AEs were of mild or moderate intensity, with no relevant difference in distribution across the treatment groups. Treatment-related AEs were infrequent (11 [10.9%] for alirocumab vs 1 [1.9%] for placebo), and most were injection site reactions (3 [6.1%] for alirocumab Q2W and 2 [3.8%] for alirocumab Q4W). There were no clinically meaningful differences between alirocumab and placebo treatment in hematologic or liver parameters.

Serious AEs (SAEs) were reported by 4 patients (8.2%) receiving alirocumab vs 1 (4.0%) receiving placebo in the

Table 2. Secondary Efficacy End Points in the Intention-to-Treat Population

	Least squares mean difference in change from baseline: alicrocumab vs placebo, % (97.5% CI)			
	Q2W cohort (n = 74)	P value	Q4W cohort (n = 79)	P value
Key secondary end points^a				
LDL-C to week 12	−45.5 (−56.3 to −34.7)	<.001	−41.5 (−52.7 to −30.2)	<.001
Apolipoprotein B to week 24	−37.8 (−47.5 to −28.2)	<.001	−30.7 (−42.0 to −19.4)	<.001
Non-HDL-C to week 24	−40.7 (−52.2 to −29.1)	<.001	−31.9 (−44.1 to −19.7)	<.001
Total cholesterol to week 24	−30.8 (−39.8 to −21.9)	<.001	−23.3 (−33.5 to −13.1)	<.001
Apolipoprotein B to week 12	−38.9 (−48.2 to −29.6)	<.001	−32.8 (−42.8 to −22.7)	<.001
Non-HDL-C to week 12	−42.8 (−53.8 to −31.8)	<.001	−37.5 (−47.9 to −27.0)	<.001
Total cholesterol to week 12	−32.7 (−41.3 to −24.2)	<.001	−27.9 (−35.6 to −20.2)	<.001
Combined estimates for odds ratio (97.5% CI): alicrocumab vs placebo				
LDL-C <130 mg/dL at week 24 ^b	77.6 (6.3 to 960.0)	.001	14.9 (3.2 to 69.8)	<.001
LDL-C <130 mg/dL at week 12 ^b	26.5 (4.0 to 174.8)	<.001	40.9 (5.7 to 290.9)	<.001
LDL-C <110 mg/dL at week 24	52.7 (3.5 to 804.3)	.001	43.1 (3.7 to 498.6)	<.001
LDL-C <110 mg/dL at week 12-LOCF	41.3 (6.6 to 1) ^c	<.001	104.8 (5.2 to 2095.9)	<.001
Adjusted mean difference (97.5% CI) in change from baseline: alicrocumab vs placebo, %				
Lipoprotein(a) to week 24	−15.2 (−30.3 to −0.1)	.02	−24.9 (−44.4 to −5.4)	.004
Lipoprotein(a) to week 12 ^b	−5.6 (−21.7 to 10.4)	.43	−13.5 (−32.7 to 5.7)	.11
HDL-C to week 24 ^d	6.4 (0.5 to 12.3)	NA	4.4 (−3.6 to 12.5)	NA
Fasting triglyceride to week 24 ^e	4.3 (−19.1 to 27.6)	NA	−19.0 (−41.5 to 3.5)	NA
Apolipoprotein A1 to week 24 ^e	1.1 (−5.9 to 8.2)	NA	8.9 (1.3 to 16.5)	NA
HDL-C to week 12 ^d	5.6 (−3.1 to 14.3)	NA	7.5 (−1.7 to 16.6)	NA
Fasting triglyceride to week 12 ^e	−8.7 (−28.8 to 11.4)	NA	−8.1 (−31.3 to 15.1)	NA
Apolipoprotein A1 to week 12 ^d	−1.6 (−7.4 to 4.1)	NA	5.7 (−2.7 to 14.1)	NA

Abbreviations: HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; Q2W, dosing every 2 weeks; Q4W, dosing every 4 weeks; NA, not applicable.

^a Regarding secondary end points, the proportion of patients reaching 50% or greater reduction in LDL-C at week 24 was 21.6% in the alicrocumab Q2W cohort (vs 0% in placebo) and 32.4% in the alicrocumab Q4W cohort (vs 9.1% in placebo). At week 12, the proportion of patients reaching 50% or greater reduction in LDL-C was 25.2% in the alicrocumab Q2W cohort (vs 0% in placebo) and 31.9% in the alicrocumab Q4W cohort (vs 0.1% in placebo).

^b At the time this study was initiated, the pediatric LDL-C recommendation was <130 mg/dL. Continuous secondary efficacy end points anticipated to have a normal distribution were analyzed using the same approach as the primary end point; those anticipated to have a nonnormal distribution were analyzed

using a multiple imputation approach followed by a robust regression model using M-estimation with treatment group and randomization strata as main effects and corresponding baseline value as covariate. Categorical secondary efficacy end points were analyzed using a multiple imputation approach followed by logistic regression model adjusting for treatment group as main effect and corresponding baseline value as covariate, stratified by randomization factors.

^c Exact odds ratio vs placebo; hierarchical testing procedure did not establish significance.

^d Least squares mean difference.

^e Combined estimate for adjusted mean difference.

Q2W cohort and 2 patients (3.8%) receiving alicrocumab vs 1 (3.7%) receiving placebo in Q4W. Most SAEs were reported as resolved or recovered at the end of the double-blind period. The 2 SAEs in patients receiving alicrocumab Q4W (syncope for both) were considered treatment related by the investigator. In one of these patients, syncope led to treatment discontinuation, and an additional patient receiving alicrocumab Q4W discontinued due to a nonserious disturbance in attention and memory. No patients discontinued alicrocumab Q2W due to AEs.

Developmental parameters, including height, weight, Cogstate score, and Tanner stage, were similar between the alicrocumab and placebo groups in both cohorts (Table 3). During the open-label period, the types and rates of AEs were consistent with those reported during the double-blind period. No patients developed antialirocumab antibodies with neutralizing status, and no deaths were reported.

Discussion

In this study of pediatric patients aged 8 to 17 years with HeFH, 2 dosing regimens of alicrocumab were associated with significantly reduced LDL-C and other proatherogenic lipid parameters compared with placebo after 24 weeks of treatment. Reduction in LDL-C was achieved by first assessment after 8 weeks of alicrocumab treatment for both dosing regimens, and improvements were maintained at week 24 and week 104. Alirocumab was well tolerated with no safety concerns.

As HeFH is an inherited disorder, early intervention to control LDL-C is crucial to mitigate disease burden later in life. Statins are the preferred treatment for pediatric HeFH management; statins effectively reduce LDL-C concentrations, cardiovascular disease risk, and cardiovascular events.^{3-5,16} In the short term, statins are effective and safe in pediatric

Table 3. Safety End Points in the Safety Population

Adverse event	No. (%)			
	Q2W cohort (n = 74)		Q4W cohort (n = 79)	
	Alirocumab (n = 49)	Placebo (n = 25)	Alirocumab (n = 52)	Placebo (n = 27)
Any adverse event ^a	26.0 (53.1)	13.0 (52.0)	26.0 (50.0)	16.0 (59.3)
Most common adverse events ^a				
Laryngitis	0	0	1.0 (1.9)	0
Nasopharyngitis	7.0 (14.3)	2.0 (8.0)	1.0 (1.9)	2.0 (7.4)
Pharyngitis	1.0 (2.0)	0	0	1.0 (3.7)
Rhinitis	1.0 (2.0)	1.0 (4.0)	1.0 (1.9)	0
Sinusitis	1.0 (2.0)	0	0	0
Tonsillitis	3.0 (6.1)	1.0 (4.0)	0	1.0 (3.7)
Upper respiratory tract infection	3.0 (6.1)	3.0 (12.0)	3.0 (5.8)	3.0 (11.1)
Headache	3.0 (6.1)	2.0 (8.0)	4.0 (7.0)	1.0 (3.7)
Migraine	0	2.0 (8.0)	0	0
Adverse events of special interest				
Injection site reaction	3.0 (6.1)	0	2.0 (3.8)	0
General allergic reaction	0	0	2.0 (3.8)	0
Disturbance in attention and memory	0	0	1.0 (1.9)	0
Hypoesthesia	0	0	1.0 (1.9)	0
Adverse events leading to discontinuation				
Disturbance in attention and memory	0	0	1.0 (1.9)	0
Syncope	0	0	1.0 (1.9)	0
Serious adverse event				
Appendicitis	0	1.0 (4.0)	0	0
Depression	1.0 (2.0)	0	0	0
Abdominal hernia	1.0 (2.0)	0	0	0
Abdominal pain	1.0 (2.0)	0	0	0
Sympathetic posterior cervical syndrome	1.0 (2.0)	0	0	0
Syncope	0	0	2.0 (3.8)	0
Noncardiac chest pain	0	0	0	1.0 (3.7)
Treatment-related adverse event	4.0 (8.2)	1.0 (4.0)	7.0 (13.5)	0
Developmental parameters				
Height change from baseline at week 24, mean (SD), cm	2.73 (4.15)	2.13 (2.28)	1.68 (2.11)	1.75 (1.27)
Weight change from baseline at week 24, mean (SD), kg	1.71 (2.47)	1.89 (2.91)	1.66 (3.24)	2.15 (2.86)

(continued)

patients, and there is some evidence that the risk of longer-term AEs is low.⁴ However, a proportion of patients remain unable to achieve LDL-C targets with statins alone.¹⁷⁻¹⁹

Table 3. Safety End Points in the Safety Population (continued)

Adverse event	No. (%)			
	Q2W cohort (n = 74)		Q4W cohort (n = 79)	
	Alirocumab (n = 49)	Placebo (n = 25)	Alirocumab (n = 52)	Placebo (n = 27)
Cogstate Battery test change from baseline at week 24, mean (SD)				
Detection test	-0.05 (0.07)	-0.04 (0.13)	-0.04 (0.08)	-0.04 (0.06)
Identification test	-0.05 (0.07)	-0.04 (0.08)	-0.05 (0.08)	-0.02 (0.13)
One Card Learning test	0.03 (0.12)	-0.02 (0.24)	-0.06 (0.11)	0.04 (0.11)
Groton Maze Learning test	4.04 (14.57)	-13.13 (29.83)	-0.31 (24.75)	5.19 (16.37)
Composite score	-0.313 (0.444)	-0.403 (1.008)	-0.136 (0.637)	-0.218 (0.501)
Tanner stage, boys				
Prepubescent				
Baseline	4 (21.1)	1 (5.9)	0	5 (41.7)
Week 24	3 (17.6)	0	0	1 (9.1)
Pubescent				
Baseline	13 (68.4)	13 (76.5)	14 (77.8)	4 (33.3)
Week 24	11 (64.7)	13 (76.5)	12 (70.6)	7 (63.6)
Postpubescent				
Baseline	2 (10.5)	3 (17.6)	4 (22.2)	3 (25.0)
Week 24	3 (17.6)	4 (23.5)	5 (29.4)	3 (27.3)
Tanner stage, girls				
Prepubescent				
Baseline	4 (13.3)	1 (12.5)	7 (20.6)	1 (6.7)
Week 24	4 (13.3)	1 (12.5)	2 (7.4)	1 (8.3)
Pubescent				
Baseline	16 (53.3)	6 (75.0)	13 (38.2)	8 (53.3)
Week 24	15 (53.6)	5 (62.5)	16 (59.3)	6 (50.0)
Postpubescent				
Baseline	10 (33.3)	1 (12.5)	14 (41.2)	6 (40.0)
Week 24	9 (32.1)	2 (25.0)	9 (33.3)	5 (41.7)
hs-CRP, mean (SD), mg/L				
Baseline	2.4 (7.9)	0.6 (0.8)	1.7 (3.4)	0.4 (0.3)
Week 24	1.1 (1.6)	1.2 (2.1)	1.8 (4.6)	0.6 (0.6)

Abbreviations: hs-CRP, high-sensitivity C-reactive protein; Q2W, dosing every 2 weeks; Q4W, dosing every 4 weeks.

^a Treatment-emergent adverse events with high-level term $\geq 5\%$ in any treatment group in the safety population.

Additional LLTs are needed for pediatric patients who cannot meet LDL-C targets with statins alone. Recent therapeutic developments include anti-PCSK9 compounds, such as the monoclonal antibodies alirocumab⁸ and evolocumab⁹ and the small interfering ribonucleic acid, inclisiran.²⁰

The efficacy of alirocumab reported here in pediatric patients with HeFH is generally consistent with that in adult patients, although the LDL-C reduction in the current study appeared to be lower than reductions previously reported in adults.¹⁰ In the Randomized, Double-Blind, Placebo-Controlled, Parallel Group Study to Evaluate the Efficacy and Safety of SAR236553/REGN727 in Patients With Heterozygous

Familial Hypercholesterolemia Not Adequately Controlled With Their Lipid-Modifying Therapy (ODYSSEY FH I) and Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Study to Evaluate the Efficacy and Safety of REGN727/SAR236553 in Patients With Heterozygous Familial Hypercholesterolemia Not Adequately Controlled With Their Lipid-Modifying Therapy (FH II),¹⁰ adults with inadequately controlled HeFH receiving alirocumab had –57.9% and –51.4% placebo-adjusted reductions in LDL-C from baseline to week 24. In the current study, the placebo-adjusted reduction from baseline to week 24 was –43.3% in the Q2W cohort and –33.8% in Q4W. Similar lipid-lowering disparities between adults and pediatric patients with HeFH have been observed with evolocumab, indicated in adult^{21,22} and pediatric HeFH.⁹ In the phase 3 Double-blind, Randomized, Placebo-controlled, Multicenter Study to Evaluate Safety, Tolerability and Efficacy of AMG 145 on LDL-C in Subjects With Heterozygous Familial Hypercholesterolemia (RUTHERFORD-2)²² of adults with HeFH, absolute LDL-C change at week 12 was –59.2% and –61.3% for Q2W and Q4W dosing, whereas in the phase 3 Double-blind, Randomized, Multicenter, Placebo-Controlled Study to Characterize the Efficacy, Safety, and Tolerability of 24 Weeks of Evolocumab for LDL-C Reduction in Pediatric Subjects 10 to 17 Years of Age With HeFH (HAUSER)⁹ of pediatric patients the placebo-adjusted LDL-C change was –38.3% at week 24 with evolocumab Q4W (absolute reduction, –44.5%). During the open-label extension of HAUSER, this reduction was maintained, at week 80 the mean percentage change in LDL-C was –35.3%.²³ It is unclear if there are clinical factors that might explain this apparent difference in response to PCSK9 inhibitors between pediatric and adult patients with HeFH. Concentrations of free PCSK9 can differ between pediatric and adult patients.²⁴ However, it has been shown that free PCSK9 concentrations do not appear to predict alirocumab response⁸; therefore, it is unlikely that this explains the lipid-lowering disparity observed between adult and pediatric populations. Others have suggested that differences in patient populations related to ezetimibe intolerance may be associated with alirocumab response¹²; however, there was relatively low use of ezetimibe in this study compared with FH I and FH II (56.0% and 67.1%, respectively).¹⁰ Regardless, comparisons across clinical trials should be considered cautiously given differences in study design and patient populations, particularly given the differences in treatment targets and up-titration thresholds between adult and pediatric patients.^{9,10,21,22}

Pediatric guidelines recommend an LDL-C goal of less than 130 mg/dL or a reduction of 50% or more.^{7,25} In this study, more than 75% of participants achieved LDL-C less than 130 mg/dL and nearly one-third achieved 50% or greater reduction with alirocumab across both dosing cohorts. Additionally, significant improvements were observed with alirocumab in other lipid parameters, including a significant reduction in Lp(a) at week 24. The HAUSER study showed a modest, nonstatistically

significant reduction in Lp(a) with evolocumab.⁹ Additionally, significant reductions were seen at weeks 12 and 24 in both non-HDL-C and apo B, established biomarkers of cardiovascular risk.^{26,27}

Alirocumab was generally well tolerated, with a safety profile consistent with previous studies in pediatric and adult patients with HeFH.^{8,10} There were 2 incidences of syncope, 1 of which led to discontinuation. Both cognitive and physical development were consistent with those expected for this age group, in line with previous developmental progression of pediatric patients in HAUSER,^{9,23} and neurocognitive AEs were rare and did not appear to differ from placebo.

Limitations

The strengths of this study include the double-blind, randomized trial design with minor eligibility restrictions. While the sample size was relatively small, adequate power for statistical analysis was provided and was similar to other studies investigating lipid-lowering therapies in pediatric populations with familial hypercholesterolemia.^{4,9,21} Most patients were White, potentially limiting generalizability. However, race was not a significant covariate in an analysis of alirocumab efficacy in adult patients,²⁸ so it is unlikely that race would impact efficacy in pediatric patients. Safety is a key factor for adherence to statin therapy in pediatric patients, and it is also known that increasing statin dose has limited additional efficacy. Therefore, although most patients were not receiving maximal approved statin doses nor using ezetimibe, the study population reflects that of specialized centers treating pediatric patients with HeFH.^{9,18,21} It may be noted that there was a difference between LDL-C lowering with alirocumab relative to placebo in the 2 cohorts, likely driven by the LDL-C increase for placebo in Q2W vs a decrease in Q4W at week 24. However, no direct comparisons can be made between dosing cohorts due to sequential enrollment of the Q2W and Q4W cohorts. Although reduction in LDL-C with alirocumab appeared to be moderate during the open-label period, this was likely due to dose adjustments being directed by investigators rather than required per study protocol. Furthermore, potential variability in protocol adherence or deviations is inherent to performing a study using different sites.

Conclusions

In this trial, 2 dosing regimens of alirocumab significantly reduced LDL-C in pediatric patients with HeFH inadequately controlled with statins, with continued efficacy over 2 years of treatment. Both regimens were generally well tolerated and consistent with previous studies in pediatric and adult patients with HeFH. Alirocumab presents a suitable adjunct option to stable LLT in pediatric patients as young as 8 years old with HeFH.

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